



NEVS (NATIONAL EMPLOYMENT VERIFICATION SYSTEM) :

A BLOCKCHAIN-BASED ENTERPRISE SOLUTION

Brief Description:

NEVS (National Employment Verification System) is a robust, production-ready enterprise blockchain solution built on Hyperledger Fabric. The platform offers a highly secure, multi-organization framework for recording and verifying employment records while ensuring strict cross-organizational consensus. The backend is built using Node.js and Express 4, connected via the Fabric Network Gateway, while the smart contracts (Chaincode) are written in Go leveraging the Chaincode as a Service (CCaaS) architecture. It features a dual-endorsement policy, dynamic service discovery, and built-in Private Data Collections (PDC) to guarantee data integrity, immutability, and enterprise-grade access control for sensitive data.

Principle:

The platform employs Enterprise Blockchain Technology—specifically Hyperledger Fabric—to deliver a decentralized and cryptographically secure ledger for verifying employment history. It uses a Raft consensus orderer alongside a strict dual-endorsement policy, requiring digital signatures from two independent network organizations (CentralGovt and CompanyOrg) before committing any state change. Additionally, Private Data Collections (PDC) are utilized via transient data routing over mTLS to completely isolate and secure highly sensitive attributes, such as salaries and compensation. This ensures that sensitive information is completely hidden from unauthorized peers and ordering nodes, achieving perfect regulatory compliance and data privacy.

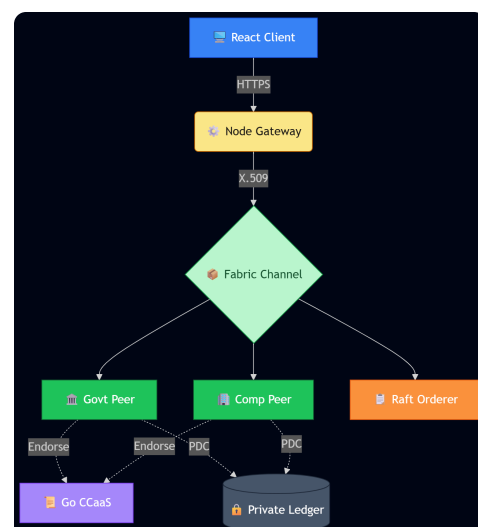
Beneficiaries:

- Government Regulatory Bodies
- Corporate HR Departments
- Employees & Job Seekers
- Background Verification Agencies
- Auditing & Compliance Firms

Benefits:

- Immutable and tamper-proof employment records
- Decentralized trust via multi-org consensus
- Enhanced data privacy via Private Data Collections
- High-throughput scalable enterprise architecture

SYSTEM ARCHITECTURE FLOW



- Automated routing & dynamic service discovery
- Secure integrations via REST API Gateway

Technologies Used:

- Hyperledger Fabric (Blockchain Framework)
- Node.js 20 + Express 4 (API Gateway)
- Go / Golang (Smart Contracts)
- Docker & Docker Compose (Containerization)
- LevelDB (Peer State Database)
- React.js + TypeScript (Frontend)

Features:

- Cross-Organizational Consensus
- Private Data Management (PDC)
- Chaincode as a Service (CCaaS)
- REST API Gateway Integration
- Dynamic Endorsement Discovery
- Dual-Endorsement Policy

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